CHAPTER 1

# Sequences and discrete models

**Key point · Chapter path.** **Objectives.** Turn a story into a sequence; distinguish increment from rate; compute terms and sums; justify monotonicity; investigate a threshold.

**Prerequisites.** Algebraic manipulation, integer powers, percentages, affine function, Python loop.

**Yearly reference point.** 8–17 September: 8 hours, then revisits

## Diagnostic: six questions, 10 minutes

Answer without consulting the lesson. Mark the questions for which help is needed. This diagnostic is not graded.

1. Calculate $3 + 2 \times 4$.
2. Calculate $2^0$, $2^3$ and $2^{-2}$.
3. A population of 80 increases by 5%. Give the new population.
4. For $f(x) = 3x - 2$, calculate $f(0)$ and $f(4)$.
5. How many integers are there from 3 to 10, endpoints included?
6. Which values does the loop `for k in range(4)` give to $k$?

**After the diagnostic.** Revisit the flagged topics with the teacher; redo a similar question, then explain the rule used.

**Speaking and logic.** In 90 seconds, explain the difference between index, term and number of terms. Give a counterexample to a rule guessed from three values.

## Investigate before formalising

In pairs, compare two collection offers: 20 kg in the first week then 5 kg more each week; 20 kg then a 20% increase each week. Conjecture which collects more in total over six weeks, then justify. Allow 10 min for investigation and 5 min for comparison.

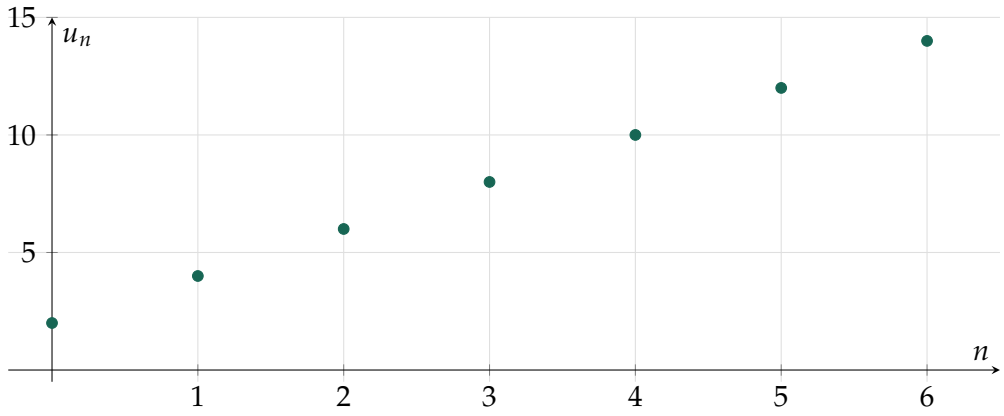
**Organisation.** Individual investigation 5 min, comparison in pairs 5 min, class discussion 5 min. Keep a record of one attempt, even an unfinished one. A conjecture must then be justified.

## 1.1 Defining and representing a sequence

**Key point · Definition and results.** A real sequence is a function whose variable is an integer: to each allowed index $n$ corresponds a real number $u_n$. The notation $(u_n)$ denotes the sequence, $u_n$ a term. An explicit formula gives $u_n$ directly in terms of $n$. A definition by recurrence gives a first term and a rule for passing from one term to the next. The representation is a set of points $(n, u_n)$: these points are not necessarily joined up.

**Status and justification.** Definition. The uniqueness of the terms obtained from a first term and a deterministic rule is accepted; no formal proof by induction is required.

**Example 1.** For $u_n = n^2 + 1$, $u_0 = 1$, $u_3 = 10$. For $v_0 = 1$ and $v_{n+1} = 2v_n + 1$, $v_1 = 3$, $v_2 = 7$, $v_3 = 15$. We cannot replace $v_3$ by $2 \times 3 + 1$: the rule applies to the previous term.




Terms of $u_n = 2n + 2$: only integer indices are plotted.

**Watch out.** The index is not the value. A finite list of terms does not determine a unique rule.

**Application.** Calculate the first four terms of $w_0 = 2$, $w_{n+1} = w_n + 2n + 1$; suggest an explicit formula.

## 1.2 Arithmetic sequences

**Key point · Definition and results.** A sequence is arithmetic with common difference $r$ if $u_{n+1} = u_n + r$ for every allowed index. Then $u_n = u_p + (n - p)r$ for $n \geq p$. If $r > 0$ it is strictly increasing; if $r < 0$, strictly decreasing; if $r = 0$, constant. The points lie on a straight line.

**Status and justification.** For fixed $p$, the sequence $v_n = u_p + (n - p)r$ satisfies $v_p = u_p$ and $v_{n+1} = v_n + r$. By uniqueness of the successive terms, $u_n = v_n$. Monotonicity follows from the sign of $u_{n+1} - u_n = r$.

**Example 2.** A tank holds 140 L and receives 12 L per minute. After $n$ minutes, $u_n = 140 + 12n$. At 5 min: 200 L. If we start the indexing at 1 with 140 L, we write $u_n = 140 + 12(n - 1)$.

**Watch out.** Adding a constant percentage does not generally give an arithmetic sequence.

**Application.** We know $u_3 = 17$ and $u_8 = 37$ for an arithmetic sequence. Determine $r$ and $u_0$.

## 1.3 Geometric sequences

**Key point · Definition and results.** A sequence is geometric with common ratio $q$ if $u_{n+1} = qu_n$. For $n \geq p$, $u_n = u_p q^{n-p}$, with the convention that an empty product equals 1. A constant rate $t$ gives $q = 1 + t$. Monotonicity is determined using $u_{n+1} - u_n = (q-1)u_n$. For $q < 0$, the signs may alternate.

**Status and justification.** The formula gives the first term, and multiplying it by $q$ gives the next term. The uniqueness of the generation justifies the formula. For $q = 0$, all terms after the first are zero; treat this case directly.

**Example 3.** A quantity of 160 mg loses 25% at each step: $u_n = 160(\frac{3}{4})^n$. The terms are positive and $u_{n+1} - u_n = -\frac{u_n}{4} < 0$. We obtain $u_2 = 90$ mg, without confusing a 25% decrease with removing 25 mg.

**Watch out.** $q > 1$ is not enough to conclude that every geometric sequence is increasing: the sign of the first term matters.

**Application.** Study the monotonicity of $u_n = -2 \times 3^n$, then of $v_n = 2(-1)^n$.

## 1.4 Sums, indices and proofs

**Key point · Definition and results.** For $n \geq 1$, $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$. For $q \neq 1$ and $n \geq 0$, $1 + q + \cdots + q^n = \frac{1 - q^{n+1}}{1 - q}$. For $q = 1$, the sum equals $n + 1$. Hence, from $u_p$ to $u_n$, an arithmetic sum equals $\frac{(n - p + 1)(u_p + u_n)}{2}$; a geometric sum equals $\frac{u_p(1 - q^{n-p+1})}{1 - q}$ if $q \neq 1$.

**Status and justification.** Writing $S = 1 + \cdots + n$ in both directions, each pair in the sum $2S$ equals $n + 1$, and there are $n$ pairs. For $T = 1 + q + \cdots + q^n$, subtract $qT = q + \cdots + q^{n+1}$: $(1 - q)T = 1 - q^{n+1}$. The case $q = 1$ must come before the division.

**Example 4.** For $u_n = 3 + 2n$, $u_2 + \cdots + u_6 = \frac{5(7 + 15)}{2} = 55$. For $v_n = 2 \times 3^n$, $v_0 + \cdots + v_3 = \frac{2(1 - 3^4)}{1 - 3} = 80$.

**Watch out.** Count the terms before using the formula: from 0 to n there are n+1.

**Application.** Calculate $\sum_{k=2}^{5} 2^k$ without the summation symbol in the answer.

## 1.5 Conjecturing a limit and programming a threshold

**Key point · Definition and results.** A limit is approached here intuitively. The terms $\dfrac{1}{n+1}$ get closer to 0; $n^2$ grows without bound; $(-1)^n$ does not settle. A numerical experiment suggests a conclusion but does not prove a behaviour for all indices. A threshold algorithm must specify the direction of the inequality and provide a stopping condition.

**Status and justification.** Illustrative observations, without a formal theory of limits. The correctness of an integer threshold is checked by examining the index found and the previous index, once monotonicity is established.

**Example 5.** For $u_n = 80(\frac{1}{2})^n$, the first index with $u_n < 6$ is 4: $u_3 = 10 \geq 6$ and $u_4 = 5 < 6$. The program starts with $n = 0, u = 80$ and repeats the division by 2 while $u \geq 6$.

**Watch out.** A loop with no bound may never stop if the threshold cannot be reached; $<$ and $\leq$ are not interchangeable.

**Application.** For $u_n = 48(\frac{1}{2})^n$, determine the first indices where $u_n \leq 6$ and where $u_n < 6$.

## Exercise path

Start with exercises 1 and 2. Then choose one of exercises 3 or 4 and do it without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher designates one. Exercise 8 is revisited after a delay.

### Exercise 1.1 · Getting to grips with indices

Consider the sequence $(u_n)$ defined for every natural number $n$ by $u_n = 2n^2 - 3n$. The first term is therefore the one of index 0.

1. Calculate $u_0$, $u_1$ and $u_4$ separately, showing the substitution of $n$.
2. Express $u_{n+1}$ in terms of $n$, then expand and simplify this expression.
3. Calculate $u_{n+1} - u_n$. By studying its sign for integer indices, describe the monotonicity of the sequence.

### Exercise 1.2 · Choosing the right model

A nursery currently has 100 young plants. It is considering two models to forecast its stock: in model A, the stock increases by 15 plants per month; in model B, it increases by 10 percent per month. Let $a_n$ and $b_n$ be the forecast stocks after $n$ months, with $a_0 = b_0 = 100$. The two models are studied separately.

1. **Model A.** Write the relation between $a_{n+1}$ and $a_n$. Recognise the type of sequence, then express $a_n$ in terms of $n$. Calculate $a_2$.
2. **Model B, after studying geometric sequences.** Answer the same questions for $(b_n)$ and calculate $b_2$.
3. For each model, explain what is assumed to be constant. Give one reason why a very long-term forecast might become unrealistic.

### Exercise 1.3 · Recovering a sequence

The sequence $(u_n)$ is geometric, defined from index 0. Its common ratio $q$ is strictly positive. We only know $u_1 = 6$ and $u_3 = 54$.

1. Determine $q$ by relating the two known terms. Present all the steps that lead to choosing its value.
2. Deduce $u_0$, then $u_5$.
3. What would change if the hypothesis $q > 0$ were removed? Give all the possible common ratios in that case and explain whether the value of $u_5$ would be affected.

### Exercise 1.4 · Rows of seats

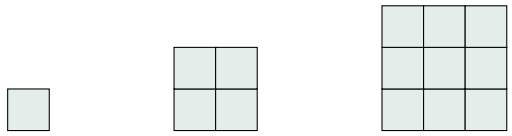
A theatre has 12 rows. The first row has 18 seats; each following row has two more seats than the previous one. Let $u_n$ be the number of seats in row $n$, for $1 \leq n \leq 12$.

1. Define the sequence $(u_n)$ and calculate the number of seats in the last row.
2. Calculate the total capacity of the theatre. Explain the number of terms used in the sum.
3. Calculate the average number of seats per row and use this result to check the total found.

### Exercise 1.5 · Accumulating squares

Patterns are built with identical square tiles. The pattern of index 1 contains one tile. To go from the pattern of index $n$ to that of index $n+1$, we add $2n+1$ tiles. Let $u_n$ be the number of tiles in the pattern of index $n$, for $n \geq 1$.

1. Calculate $u_2$ and $u_3$. Draw an arrangement of these tiles that suggests a formula for $u_n$.
2. Conjecture an expression for $u_n$ in terms of $n$.
3. Check that this expression gives the first term and satisfies the construction rule. Explain why these two checks identify the sequence.




One possible arrangement of the first three patterns; the drawing helps to state a conjecture, which still has to be justified.

### Exercise 1.6 · Two repayments

A person must repay 600 monetary units, with no interest. Two plans are considered. With plan A, they pay 50 units at the end of each month until the debt is repaid. With plan B, they pay each month half of what they still owe. The calculations for model B are carried out without rounding. Let $a_n$ and $b_n$ be the remaining debts after the payment of month $n$, with $a_0 = b_0 = 600$.

1. Express $a_n$ until the debt is cleared. In which month is the repayment completed?
2. Express $b_n$. Determine the first month in which the remaining debt is strictly less than 10 units; check the previous month.
3. In model B without rounding, does the debt become exactly zero after a finite number of months? Justify and comment on the difference with a real situation.

### Exercise 1.7 · Reading a list

A program must keep the terms of the sequence defined by $u_0 = 2$ and $u_{n+1} = 3u_n - 1$. It starts with the instructions `u=2` and `L=[u]`. It then executes five times, in this order, `u=3*u-1` then `L.append(u)`.

1. Execute the first two passes of the loop by hand. Then give the contents of the list.
2. How many elements does the list contain at the end of the five passes? What is the index of the last term stored?
3. Write the complete program in Python, then check the value of the last element. Explain why the number of terms exceeds the number of passes by one.

### Exercise 1.8 · Delayed review

This question is done one week after the chapter, without consulting the formulas. A sequence is defined for $n \geq 2$ by $u_2 = 12$ and $u_{n+1} = u_n - 3$.

1. Calculate $u_7$, stating how many passes separate the indices 2 and 7.
2. Calculate the sum $u_2 + u_3 + \cdots + u_7$. Explain the difference between the number of terms in this sum and the number of passes calculated in the first question.

**Homework.** Do only the task designated at the end of the session, in the dated progression: 10–20 min at most. The exercises depend on the topics already studied, so some questions are deferred. Revisit exercise 8 one week after the chapter (10 min). If the allotted time is exceeded, note the difficulty and bring your attempts; do not pile up the whole reserve.

## To do in the CQFD textbook

These exercises are done in your textbook. The progression indicates the session; they replace the homework planned for that day (10 to 20 minutes in total).

**CQFD, exercise 9 p. 64, questions a and b.** Calculate the first four terms. Write the substitutions before simplifying.

**CQFD, exercise 11 p. 64, all questions.** Calculate the terms one after another and keep a record of the four steps.

## MCQ 1: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $u_n = 3n + 2$: $u_4$ equals
   - A. 14
   - B. 18
   - C. 12
   - D. 6
2. $u_0 = 2$, $u_{n+1} = u_n + 5$: $u_3$ equals
   - A. 10
   - B. 12
   - C. 17
   - D. 15
3. An increase of 8% gives the ratio
   - A. 0.08
   - B. 1.8
   - C. 0.92
   - D. 1.08
4. From $u_3$ to $u_9$ inclusive, there are
   - A. 6
   - B. 7
   - C. 9
   - D. 12
5. $1 + 2 + \cdots + 20$ equals
   - A. 200
   - B. 420
   - C. 210
   - D. 190

## MCQ 2: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $u_n = -3 \times 2^n$ is
   - A. strictly increasing
   - B. constant
   - C. neither increasing nor decreasing
   - D. strictly decreasing
2. $1 + 2 + 4 + 8$ equals
   - A. 15
   - B. 16

C. 14
D. 8

3. For $u_n = \dfrac{16}{2^n}$, the first index where $u_n < 2$:
   A. 2
   B. 3
   C. 4
   D. 5
4. The sequence $(-1)^n$
   A. tends to 0
   B. has no limit
   C. is increasing
   D. is constant
5. A list contains $u_0$ to $u_{10}$; its length is
   A. 10
   B. 9
   C. 12
   D. 11

## True or false? Justify.

1. Every geometric sequence with common ratio $q > 1$ is increasing.
2. If two sequences have their first three terms equal, they are equal.
3. For $q = 1$, $1 + q + \cdots + q^n = n + 1$.

**Key point · Self-assessment.** I distinguish explicit from recurrence definitions; I recognise a constant increment or rate; I count the terms; I check the previous index of a threshold.

For each objective, indicate: achieved without help; achieved with help; to be reworked. Write one example and one mistake to avoid.

# Solutions and teaching notes for chapter 1

**Teacher use.** The answers below concern the tasks not worked in the lesson. First have students state their method; give graded help before the solution. The durations are estimates to be calibrated in class.

## Diagnostic and remediation

1. 11: multiplication comes before addition. *Action:* Rewrite the steps of two calculations with brackets.
2. 1, 8, $\frac{1}{4}$. *Action:* Build the chain of powers by dividing by 2.
3. $80 \times 1.05 = 84$. *Action:* Link 5% to $\frac{5}{100}$ and then to the coefficient 1.05.
4. $-2$ and 10. *Action:* Substitute x in brackets before calculating.
5. $10 - 3 + 1 = 8$. *Action:* Write out the list, then recover the calculation.
6. $0, 1, 2, 3$. *Action:* Draw a two-column table, index and value; test range(1,4).

## Investigation activity

Offer A: last week 45 kg, sum $\frac{6(20+45)}{2} = 195$ kg. B: $\frac{20(1-1.2^6)}{1-1.2} = 198.5984$ kg. B exceeds A in total, even though some of its early terms are smaller. A calculator is allowed here; argue from the cumulative total and not from the last term alone.

## Immediate applications

1. $2, 3, 6, 11$. The formula $w_n = n^2 + 2$ works: it gives the first term and $(n+1)^2 + 2 - (n^2+2) = 2n + 1$.
2. $37 - 17 = 5r$, so $r = 4$ and then $u_0 = 17 - 3 \times 4 = 5$.
3. $u_{n+1} - u_n = -4 \times 3^n < 0$: decreasing. $v_n$ alternates between 2 and $-2$: neither increasing nor decreasing.
4. $4 + 8 + 16 + 32 = 60$, or $\frac{4(1-2^4)}{1-2} = 60$.
5. $u_3 = 6$: index 3 for $\leq 6$, index 4 for $< 6$.

## Exercises: method and results

**Exercise 1.1 – Getting to grips with indices.** $u_0 = 0, u_1 = -1, u_4 = 20$. Expand with brackets: $u_{n+1} = 2(n+1)^2 - 3(n+1) = 2n^2 + n - 1$; difference $4n - 1$. The sign changes between indices 0 and 1: the sequence first decreases, then increases from index 1 onwards.

**Exercise 1.2 – Choosing the right model.** $a_n = 100 + 15n$ and $b_n = 100(1.1)^n$, so $a_2 = 130$, $b_2 = 121$. A assumes a constant net addition; B a constant net rate. In reality, rounding and capacity limit these models.

**Exercise 1.3 – Recovering a sequence.** $u_3 = q^2 u_1$ so $q^2 = 9$. Positivity forces $q = 3$, hence $u_0 = 2$ and $u_5 = 486$. Without this condition, $q = -3$ would also be possible, with $u_0 = -2$ and the same $u_5 = 486$.

**Exercise 1.4 – Rows of seats.** The last row has $18 + 11 \times 2 = 40$ seats. Sum $\dfrac{12(18+40)}{2} = 348$. Check: average 29 seats per row, between 18 and 40.

**Exercise 1.5 – Accumulating squares.** We obtain 4 then 9 squares. A square of integer side n completed by a row and a column needs $2n+1$ cells. The formula $n^2$ has the right first term and $(n+1)^2 - n^2 = 2n+1$. The generation is the same.

**Exercise 1.6 – Two repayments.** For A: $a_n = 600 - 50n$ for $0 \leq n \leq 12$, then 0. For B: $b_n = \dfrac{600}{2^n}$. $b_5 = 18.75$ and $b_6 = 9.375$; strict threshold at index 6. B never gives exactly 0 in this ideal model: do not announce a finite repayment.

**Exercise 1.7 – Reading a list.** The first three elements are 2, 5, 14. An initial value and five additions give six elements; indices 0 to 5. The last one is $u_5 = 365$.

**Exercise 1.8 – Delayed review.** $u_7 = 12 + 5(-3) = -3$. Six terms: $\dfrac{6(12-3)}{2} = 27$. The 5 counts the passes, the 6 the terms.

**CQFD, exercise 9 p. 64, questions a and b.** a: $-1, 0, 3, 8$. b: $2, \dfrac{7}{2}, 4, \dfrac{17}{4}$. The first index is 0; the denominator of b is $n+1$ and never vanishes on $\mathbb{N}$.

**CQFD, exercise 11 p. 64, all questions.** Starting from $u_0 = 2$, the rule $u_{n+1} = 2u_n - 1$ gives $u_1 = 3$, $u_2 = 5$, $u_3 = 9$, $u_4 = 17$. The rule uses the previous term, not its index.

## MCQ: answers and reasons

**MCQ 1.**

1. A. $3 \times 4 + 2 = 14$.
2. C. Three additions: $2 + 3 \times 5 = 17$.
3. D. The coefficient is $1 + \dfrac{8}{100}$.
4. B. $9 - 3 + 1 = 7$.
5. C. $\dfrac{20 \times 21}{2} = 210$.

**MCQ 2.**

1. D. The difference is $-3 \times 2^n < 0$.
2. A. Four terms; $\dfrac{1 - 2^4}{1 - 2} = 15$.
3. C. $u_3 = 2$, $u_4 = 1$.

4. B. The terms alternate between two distinct values.
5. D. Eleven indices, from 0 to 10.

### True or false

1. False: $u_n = -2^n$ is strictly decreasing.
2. False: $u_n = 0$ and $v_n = n(n-1)(n-2)$ coincide at indices 0, 1, 2, but not at index 3.
3. True: each term equals 1 and there are n+1 terms.

**Differentiated teaching.** For a technical block, offer only the first step (domain, factor or reference sample space), then withdraw the help. For an independent student, ask for a second method and the treatment of a limiting case. Orally, allow an initial reformulation in the language the student has mastered, then require a precise mathematical sentence in French.